EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Let $P(x) = kx^3 + 2k^2x^2 + k^3$. The sum of all real numbers k for which $(x-2)$ is a factor of $P(x)$, is

(A) 4 (B) 8 (C) -4 (D) -8

MFN001

2. Let $P(x) = x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f$ be a polynomial such that $P(1) = 1; P(2) = 2; P(3) = 3; P(4) = 4; P(5) = 5$ and $P(6) = 6$, then find the value of $P(7)$.

(A) 700 (B) 727 (C) 650 (D) 725

MFN002

3. The domain of definition of the function, $y(x)$ is given by the equation, $2^x + 2^y = 2$ is -

(A) $0 < x \le 1$ (B) $0 \le x \le 1$ (C) $-\infty < x \le 0$ (D) $-\infty < x < 1$

MFN003

4. Range of function $f(x) = \log_2\left(\frac{4}{\sqrt{x+2}+\sqrt{2-x}}\right)$ is given by

(A) $(0,\infty)$ (B) $\left[\frac{1}{2},1\right]$ (C) $[1,2]$ (D) $\left[\frac{1}{4},1\right]$

MFN004

5. Range of $f(x) = \frac{\sec x + \tan x - 1}{\tan x - \sec x + 1}; x \in \left(0, \frac{\pi}{2}\right)$ is-

(A) (0,1) (B) $(1,\infty)$ (C) $(-1,0)$ (D) $(-\infty,-1)$

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MFN005

6. Let $f(x) = \frac{x}{1-x}$ and let α be a real number. If $x_0 = \alpha, x_1 = f(x_0), x_2 = f(x_1), \ldots\ldots$ & $x_{2011} = -\frac{1}{2012}$ then the value of α is -

(A) $\frac{2011}{2012}$ (B) 1 (C) 2011 (D) -1

MFN006

7. If $f_1(x) = 2^{f_2(x)}$, where $f_2(x) = 2012^{f_3(x)}$, where $f_3(x) = \left(\frac{1}{2013}\right)^{f_4(x)}$, where $f_4(x) = \log_{2013}\log_x 2012$, then the range of $f_1(x)$ is -

(A) $(2,\infty)$ (B) $(2012,\infty)$ (C) $(0,\infty)$ (D) $(-\infty,\infty)$

MFN007

8. Let $f: \mathbb{R} - \left\{\frac{-15}{2}\right\} \to \mathbb{R} - \left\{\frac{1}{2}\right\}$ be defined by $f(x) = \frac{x+10}{2x+15}$ then $f(x)$ is-

(A) one-one but not onto (B) many one but not-onto
(C) one-one and onto (D) many one and onto

MFN008

9. $f: \mathbb{R} \to \mathbb{R}$ $f(x) = \frac{2x^2-5x+3}{8x^2+9x+11}$, then f is -

(A) one-one onto (B) many-one onto (C) one-one into (D) many one into

MFN009

For More Join: @IITJEE_Advanced

10. If functions $f(x)$ and $g(x)$ are defined on $\mathbb{R}\to\mathbb{R}$ such that $f(x)=\begin{cases} x+3 & , \quad x\in \text{rational} \\ 4x & , \quad x\in \text{irrational} \end{cases}$,

$g(x)=\begin{cases} x+\sqrt{5} & , \quad x\in \text{irrational} \\ -x & , \quad x\in \text{rational} \end{cases}$ then $(f-g)(x)$ is-

(A) one-one & onto
(B) neither one-one nor onto
(C) one-one but not onto
(D) onto but not one-one

MFN010

11. Let $f: X \to Y$ be a function such that $f(x)=\sqrt{x-2}+\sqrt{4-x}$, then the set of X and Y for which $f(x)$ is both injective as well as surjective, is -

(A) [2,4] and $\left[\sqrt{2},2\right]$
(B) [3,4] and $\left[\sqrt{2},2\right]$
(C) [2,4] and [1,2]
(D) [2,3] and [1,2]

MFN011

12. If $f: \mathbb{R} \to \mathbb{R}$ & $f(x)=\frac{\sin([x]\pi)}{x^2+2x+3}+2x-1+\sqrt{x(x-1)+\frac{1}{4}}$ (where $[x]$ denotes integral part of x), then $f(x)$ is -

(A) one-one but not onto
(B) one-one & onto
(C) onto but not one-one
(D) neither one-one nor onto

MFN012

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13. If the function $f:\left[\frac{7}{2},\infty\right)\to\left[-\frac{9}{4},\infty\right)$ be defined by $f(x)=x^2-7x+10$ then $f^{-1}(x)$ is-

(A) $\frac{7+\sqrt{9+4x}}{2}$ (B) $\frac{7\pm\sqrt{9+4x}}{2}$ (C) $\frac{7+\sqrt{9-4x}}{2}$ (D) $\frac{7\pm\sqrt{9-4x}}{2}$

MFN013

14. $f(x)=[x-1]+\{x\}^{[x]}, x\in(1,3)$, then $f^{-1}(x)$ is –
(where [.] denotes greatest integer function and {.} denotes fractional part function)

(A) $\begin{bmatrix} x+1 & x\in(1,2) \\ 2+\sqrt{x-1} & x\in[2,3) \end{bmatrix}$
(B) $\begin{bmatrix} x-1 & x\in(1,2) \\ 2-\sqrt{x-1} & x\in[2,3) \end{bmatrix}$
(C) $\begin{bmatrix} x-1 & x\in(0,1) \\ 2-\sqrt{x-1} & x\in[1,2) \end{bmatrix}$
(D) $\begin{bmatrix} x+1 & x\in(0,1) \\ 2+\sqrt{x-1} & x\in[1,2) \end{bmatrix}$

MFN014

15. Let $f:(0,\infty)\to(1,\infty)$ be a function such that $f(x)=1+\frac{3}{2}\sqrt{x}$ and $g(x)$ is inverse of $f(x)$ then the point where $f(x)$ and $g(x)$ intersect is

(A) $\left(\frac{1}{4},\frac{1}{4}\right)$ (B) $(4,2)$ (C) $(4,4)$ (D) Does not exist

MFN015

For More Join: @IITJEE_Advanced

16. Which of the following functions is an odd function ?

(A) $|x-2|+(x+2)\,sgn(x+2)$ (B) $\frac{1}{x(e^x-1)}+\frac{1}{2x}$

(C) $\log\left(\sin x+\sqrt{1+\sin^2 x}\right)$ (D) $e^{-4x}(e^{2x}-1)^4$

(where $sgn(x)$ denotes signum function of x)

MFN016

17. Let $f:\mathbb{R}\to\mathbb{R}$ be a real valued function such that $f(10+x)=f(10-x)\forall\, x\in\mathbb{R}$ and $f(20+x)=-f(20-x)\forall x\in\mathbb{R}$. Then which of the following statements is true -

(A) $f(x)$ is odd and periodic (B) $f(x)$ is odd and aperiodic

(C) $f(x)$ is even and periodic (D) $f(x)$ is even and aperiodic

MFN017

18. Period of $f(x)=\{x\}+\left\{x+\frac{1}{3}\right\}+\left\{x+\frac{2}{3}\right\}$ is equal to (where {.} denotes fractional part function)

(A) 1 (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

MFN018

19. Period of function $f(x)=min\{\sin x,|x|\}+\frac{x}{\pi}-\left[\frac{x}{\pi}\right]$ (where [.] denotes greatest integer function) is -

(A) $\pi/2$ (B) π (C) 2π (D) 4π

MFN019

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20. Let $f(x)=2x-\left\{\frac{x}{\pi}\right\}$ and $g(x)=\cos x$, where {.} denotes fractional part function, then period of $gof(x)$ is -

(A) $\frac{\pi}{2}$ (B) π (C) $\frac{3\pi}{2}$ (D) $\frac{\pi}{4}$

MFN020

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. Let $f(x)=\begin{cases} x^2-3x+4 & ; \quad x<3 \\ x+7 & ; \quad x\geq 3 \end{cases}$ and $g(x)=\begin{cases} x+6 & ; \quad x<4 \\ x^2+x+2 & ; \quad x\geq 4 \end{cases}$, then which of the following is/are true -

(A) $(f+g)(1)=9$ (B) $(f-g)(3.5)=1$ (C) $(fg)(0)=24$ (D) $\left(\frac{f}{g}\right)(5)=\frac{8}{3}$

MFN060

2. Let $f(x)=x^2+3x+2$, then number of solutions to -

(A) $f(|x|)=2$ is 1 (B) $f(|x|)=2$ is 3

(C) $|f(x)|=0.125$ is 4 (D) $|f(|x|)|=0.125$ is 8

MFN061

3. For each real x, let $f(x)=max\{x,x^2,x^3,x^4\}$, then $f(x)$ is -

(A) x^4 for $x\leq -1$ (B) x^2 for $-1<x\leq 0$ (C) $f\left(\frac{1}{2}\right)=\frac{1}{2}$ (D) $f\left(\frac{1}{2}\right)=\frac{1}{4}$

MFN062

4. For the function $f(x)=|x+3|-|x+1|-|x-1|+|x-3|$, identify correct option(s)

(A) Range of $f(x)$ is $(-\infty,4]$ (B) maximum value of $f(x)$ is 4

(C) $f(x)=4$ has infinite solutions (D) $f(x)=0$ has infinite solutions

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MFN063

5. Which of the following statement(s) is(are) correct ?

(A) If f is a one-one mapping from set A to A, then f is onto.

(B) If f is an onto mapping from set A to A, then f is one-one

(C) Let f and g be two functions defined from $\mathbb{R}\to\mathbb{R}$ such that gof is injective, then f must be injective.

(D) If set A contains 3 elements while set B contains 2 elements, then total number of functions from A to B is 8.

MFN064

6. Consider a function $f(x)=\frac{|x+1|}{x+1}+\frac{x-1}{|x-1|}$, then choose correct option(s) regarding function f

(A) many one function (B) odd function

(C) bounded function (D) range contain exactly three integers

MFN065

7. Let $f(x)=\begin{cases} x^2 & ; \quad 0<x<2 \\ 2x-3 & ; \quad 2\leq x<3 \\ x+2 & ; \quad x\geq 3 \end{cases}$ Then :-

(A) $f\left\{f\left(f\left(\frac{3}{2}\right)\right)\right\}=f\left(\frac{3}{2}\right)$ (B) $1+f\left\{f\left(f\left(\frac{5}{2}\right)\right)\right\}=f\left(\frac{5}{2}\right)$

(C) $f\{f(1)\}=f(1)=1$ (D) $f\{f(1)\}=2$

MFN066

For More Join: @IITJEE_Advanced

8. If $\{x\}=\frac{2}{3}$ & $\left[x+\left\{x+\left[x+\{x+.....100 \text{ times}\}\right]\right\}\right]=5$, then -

(A) $x=\frac{14}{3}$ (B) $[x]=5$ (C) $x=\frac{17}{3}$ (D) $[x]=4$

(where [.] & {.} denotes greatest integer function & fractional part function respectively)

MFN067

9. Let $f:\{1,2,3,4,5\}\to\{1,2,3,4,5\}$ is such that $f(x)$ is a one-one function satisfying following condition $f(x)=x+1$ if and only if x is even (i.e. $f(3)\neq 4, f(4)=5$ etc). Then $f^{-1}(2)$ can be-

(A) 1 (B) 3 (C) 5 (D) 2

MFN068

10. Let $f:[1,\infty)\to[3,\infty)$ be defined as $f(x)=(\log_2 x)^2+2(\log_2 x)+3$. If $f^{-1}(x)$ is inverse of $f(x)$, then

(A) $f^{-1}(x)=2^{-1+\sqrt{x-2}}$ (B) $f^{-1}(x)=2^{-1-\sqrt{x-2}}$

(C) domain of $f^{-1}(x)$ is $[2,\infty)$ (D) domain of $f^{-1}(x)$ is $[3,\infty)$

MFN069

11. If $f\left(x+\frac{1}{2}\right)+f\left(x-\frac{1}{2}\right)=f(x)\ \forall x\in\mathbb{R}$, then period of function $f(x)$ is greater than-

(A) 1 (B) $\frac{3}{2}$ (C) 2 (D) 3

MFN070

COMPREHENSION TYPE QUESTIONS

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Paragraph for Question No. 12 to 13

Let $f(x)=\begin{cases} x & ; \ x<0 \\ 1-x & ; \ x\geq 0 \end{cases}$ & $g(x)=\begin{cases} x^2 & ; \ x<-1 \\ 2x+3 & ; \ -1\leq x\leq 1 \\ x & ; \ x>1 \end{cases}$

On the basis of above information, answer the following questions :

12. Range of $f(x)$ is -

(A) $(-\infty,1]$ (B) $(-\infty,\infty)$ (C) $(-\infty,0]$ (D) $(-\infty,2]$

MFN071

13. Range of $g(f(x))$ is -

(A) $(-\infty,\infty)$ (B) $[1,3)\cup(3,\infty)$ (C) $[1,\infty)$ (D) $[0,\infty)$

MFN072

Paragraph for Question No. 14 to 15

Let $f:\mathbb{R}\to\mathbb{R}$ is a function satisfying $f(2-x)=f(2+x)$ and $f(20-x)=f(x), \forall x\in\mathbb{R}$.

On the basis of above information, answer the following questions :

14. If $f(0)=5$, then minimum possible number of values of x satisfying $f(x)=5$, for $x\in[0,170]$ is-

(A) 22 (B) 23 (C) 24 (D) 25

MFN073

15. Graph of $y=f(x)$ is symmetrical about $x=k$. Find k –

(A) 15 (B) 16 (C) 17 (D) 18

MFN074

For More Join: @IITJEE_Advanced

Paragraph for Question No. 16 to 17

Let $f:(-\infty,-2]\to[3,\infty)$ defined by $f(x)=x^2+2x+3$.

On the basis of above information, answer the following questions :

16. $f^{-1}(x)$ is equal to -

(A) $\sqrt{x+2}-1$ (B) $\sqrt{x-2}-1$ (C) $-\sqrt{x+2}-1$ (D) $-\sqrt{x-2}-1$

MFN075

17. Range of the function $g(x)=f(2x)$ is equal to -

(A) $[1,\infty)$ (B) $[2,\infty)$ (C) $[3,\infty)$ (D) $[4,\infty)$

MFN076

MATCHING LIST TYPE QUESTION

18.

	List-I $f(x)$		**List-II** **Range**
(I)	$\frac{\cos^2 x+\cos x+2}{\cos^2 x+\cos x+1}$	(P)	$\left(0,\frac{7}{3}\right]$
(II)	$\left\lvert\frac{(\sqrt{\cos x}-\sqrt{\sin x})(\sqrt{\cos x}+\sqrt{\sin x})}{3(\cos x+\sin x)}\right\rvert$	(Q)	$\left[\frac{4}{3},\frac{7}{3}\right]$
(III)	$\frac{7}{3(x^6+2x^4+3x^2+1)}$	(R)	$\left[0,\frac{1}{3}\right]$
(IV)	$\log_8(x^2+2x+2)$	(S)	$[0,\infty)$

Which of the following is correct ?

(A) I $\to$ Q; II $\to$ R; III $\to$ P; IV $\to$ S (B) I $\to$ Q; II $\to$ R; III $\to$ S; IV $\to$ P

(C) I $\to$ P; II $\to$ S; III $\to$ Q; IV $\to$ R (D) I $\to$ P; II $\to$ S; III $\to$ R; IV $\to$ Q

MFN077

MATRIX MATCH TYPE QUESTION

19.

	Column-I **Number of integers in**		**Column-II**
(A)	Domain of $f(x)=\ell n\{x\}$	(P)	0
(B)	Domain of $f(x)=\sqrt{\sec(\sin x)}+\sqrt{\left[x+\frac{1}{x}\right]}+\sqrt{10-[x]^2}$	(Q)	2
(C)	Range of $f(x)=x^2-2x+2, x\in[0,2]$	(R)	3
(D)	Range of $f(x)=\sqrt{25-[x]^2}$	(S)	less than 3
		(T)	more than 3

(where [.] and {.} denote greatest integer function and fractional part function respectively)

MFN078

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EXERCISE (S)

1. The sum of integral values of the elements in the domain of $f(x)=\sqrt{\log_{\frac{1}{2}}|3-x|}$ is

MFN021

2. Let $f(x)=\frac{9^x}{9^x+3}$ then find the value of the sum $f\left(\frac{1}{2006}\right)+f\left(\frac{2}{2006}\right)+f\left(\frac{3}{2006}\right)+.....f\left(\frac{2005}{2006}\right)$

MFN022

3. The number of integral values of x satisfying the inequality $[x-5][x-3]+2<[x-5]+2[x-3]$ (where [.] represents greatest integer function) is –

MFN023

4. Suppose, $f(x,n)=\sum_{k=1}^{n}\log_x\left(\frac{k}{x}\right)$, then the value of x satisfying the equation $f(x,10)=f(x,11)$, is

MFN024

5. If $f(x)=a\log\left(\frac{1+x}{1-x}\right)+bx^3+c\sin x+5$ and $f(\log_3 2)=4$, then $f\left(\log_3\left(\frac{1}{2}\right)\right)$ is equal to

MFN025

6. If $f(x)=px+q$ and $f(f(f(x)))=8x+21$, where p and q are real numbers, then $p+q$ equals-

MFN026

7. Let $f(x)=\frac{2x-1}{x+3}$. If $f^{-1}=\frac{ax+b}{c-x}$, then $a+b+c$ is

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MFN027

8. A function $'f'$ is defined for all real numbers and satisfies $f(2+x)=f(2-x)$ and $f(7+x)=f(7-x)$ for all real x. If $x=0$ is a root of $f(x)=0$, then find least number of roots of $f(x)=0$ for $x\in[-1000,1000]$.

MFN028

9. If $f(x,y)=\max(x,y)+\min(x,y)$ and $g(x,y)=\max(x,y)-\min(x,y)$, then the value of $f\left(g\left(-\frac{2}{3},-\frac{3}{2}\right),g(-3,-4)\right)$ is equal to $\frac{k}{6}$ then k $(k\in\mathbb{N})$ is -

MFN029

10. Let $'f'$ be an even periodic function with period $'4'$ such that $f(x)=2^x-1, 0\le x\le 2$. The number of solutions of the equation $f(x)=1$ in $[-10,20]$ are

MFN030

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EXERCISE - JEE (Main) PYQ

1. For $x \in R - \{0, 1\}$, let $f_1(x) = \frac{1}{x}$, $f_2(x) = 1 - x$ and $f_3(x) = \frac{1}{1-x}$ be three given functions. If a function, $J(x)$ satisfies $(f_2 o J o f_1)(x) = f_3(x)$ then $J(x)$ is equal to :- **[JEE (Main) 2019]**

(1) $f_3(x)$ (2) $f_1(x)$ (3) $f_2(x)$ (4) $\frac{1}{x} f_3(x)$

MFN031

2. If $f(x) = \log_e\left(\frac{1-x}{1+x}\right), |x| < 1$, then $f\left(\frac{2x}{1+x^2}\right)$ is equal to : **[JEE (Main) 2019]**

(1) $2f(x)$ (2) $2f(x^2)$ (3) $(f(x))^2$ (4) $-2f(x)$

MFN035

3. Let $\sum_{k=1}^{10} f(a+k) = 16\left(2^{10} - 1\right)$, where the function f satisfies $f(x + y) = f(x)f(y)$ for all natural numbers x, y and $f(1) = 2$. then the natural number 'a' is **[JEE (Main) 2019]**

(1) 4 (2) 3 (3) 16 (4) 2

MFN036

4. If the function $f : R - \{1, -1\} \to A$ defined by $f(x) = \frac{x^2}{1-x^2}$, is surjective, then A is equal to **[JEE (Main) 2019]**

(1) $R - [-1, 0)$ (2) $R - (-1, 0)$ (3) $R - \{-1\}$ (4) $[0, \infty)$

MFN037

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5. The domain of the definition of the function $f(x) = \frac{1}{4-x^2} + \log_{10}(x^3 - x)$ is :- **[JEE (Main) 2019]**

(1) $(1, 2) \cup (2, \infty)$ (2) $(-1, 0) \cup (1, 2) \cup (3, \infty)$
(3) $(-1, 0) \cup (1, 2) \cup (2, \infty)$ (4) $(-2, -1) \cup (-1, 0) \cup (2, \infty)$

MFN038

6. If $g(x) = x^2 + x - 1$ and $(gof)(x) = 4x^2 - 10x + 5$, then $f\left(\frac{5}{4}\right)$ is equal to **[JEE (Main) 2020]**

(1) $\frac{3}{2}$ (2) $-\frac{1}{2}$ (3) $-\frac{3}{2}$ (4) $\frac{1}{2}$

MFN039

7. Let $f : (1,3) \to R$ be a function defined by $f(x) = \frac{x[x]}{1+x^2}$, where $[x]$ denotes the greatest integer $\le x$. Then the range of f is **[JEE (Main) 2020]**

(1) $\left(\frac{3}{5}, \frac{4}{5}\right)$ (2) $\left(\frac{2}{5}, \frac{3}{5}\right] \cup \left(\frac{3}{4}, \frac{4}{5}\right)$ (3) $\left(\frac{2}{5}, \frac{4}{5}\right]$ (4) $\left(\frac{2}{5}, \frac{1}{2}\right) \cup \left[\frac{3}{5}, \frac{4}{5}\right]$

MFN040

8. Let $f : R \to R$ be such that for all $x \in R$ $(2^{1+x} + 2^{1-x})$, $f(x)$ and $(3^x + 3^{-x})$ are in A.P., then the minimum value of $f(x)$ is **[JEE (Main) 2020]**

(1) 0 (2) 3 (3) 2 (4) 4

MFN041

For More Join: @IITJEE_Advanced

9. The inverse function of $f(x)=\frac{8^{2x}-8^{-2x}}{8^{2x}+8^{-2x}}, x\in(-1,1)$, is **[JEE (Main) 2020]**

(1) $\frac{1}{4}(\log_8 e)\log_e\left(\frac{1-x}{1+x}\right)$ (2) $\frac{1}{4}\log_e\left(\frac{1-x}{1+x}\right)$

(3) $\frac{1}{4}(\log_8 e)\log_e\left(\frac{1+x}{1-x}\right)$ (4) $\frac{1}{4}\log_e\left(\frac{1+x}{1-x}\right)$

MFN042

10. Let $f: R \to R$ be a function which satisfies $f(x+y)=f(x)+f(y)\forall x,y\in R$. If $f(1)=2$ and $g(n)=\sum_{k=1}^{(n-1)} f(k), n\in \mathrm{N}$ then the value of n, for which $g(n)=20$, is : **[JEE (Main) 2020]**

(1) 5 (2) 9 (3) 20 (4) 4

MFN043

11. Let $[t]$ denote the greatest integer $\le t$. Then the equation in x, $[x]^2+2[x+2]-7=0$ has : **[JEE (Main) 2020]**

(1) no integral solution (2) exactly four integral solutions

(3) exactly two solutions (4) infinitely many solutions

MFN044

12. For a suitably chosen real constant a, let a function, $f: R-\{-a\}\to R$ be defined by $f(x)=\frac{a-x}{a+x}$.

Further suppose that for any real number $x\neq -a$ and $f(x)\neq -a$, $(fof)(x)=x$. Then $f\left(-\frac{1}{2}\right)$ is equal to : **[JEE (Main) 2020]**

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(1) $\frac{1}{3}$ (2) 3 (3) –3 (4) $-\frac{1}{3}$

MFN045

13. Suppose that a function $f: R\to R$ satisfies $f(x+y)=f(x)f(y)$ for all $x,y\in R$ and $f(1)=3$. If $\sum_{i=1}^{n} f(i)=363$, then n is equal to. **[JEE (Main) 2020]**

MFN046

14. The range of the function, $f(x)=\log_{\sqrt{5}}\left(3+\cos\left(\frac{3\pi}{4}+x\right)+\cos\left(\frac{\pi}{4}+x\right)+\cos\left(\frac{\pi}{4}-x\right)-\cos\left(\frac{3\pi}{4}-x\right)\right)$ is : **[JEE (Main) 2021]**

(1) $(0,\sqrt{5})$ (2) $[-2,2]$ (3) $\left[\frac{1}{\sqrt{5}},\sqrt{5}\right]$ (4) $[0,2]$

MFN047

15. Let $f(x)$ be a polynomial of degree 3 such that $f(k)=-\frac{2}{k}$ for $k=2,3,4,5$. Then the value of $52-10f(10)$ is equal to : **[JEE (Main) 2021]**

MFN048

16. The inverse of $y=5^{\log x}$ is : **[JEE (Main) 2021]**

(1) $x=5^{\log y}$ (2) $x=y^{\log 5}$ (3) $x=y^{\frac{1}{\log 5}}$ (4) $x=5^{\frac{1}{\log y}}$

MFN049

For More Join: @IITJEE_Advanced

17. Let $f: R-\{3\} \to R-\{1\}$ be defined by $f(x)=\frac{x-2}{x-3}$. Let $g: R \to R$ be given as $g(x)=2x-3$. Then, the sum of all the values of x for which $f^{-1}(x)+g^{-1}(x)=\frac{13}{2}$ is equal to **[JEE (Main) 2021]**

(1) 7 (2) 2 (3) 5 (4) 3

MFN050

18. Let $[x]$ denote the greatest integer $\leq x$, where $x \in \boldsymbol{R}$. If the domain of the real valued function $f(x)=\sqrt{\frac{|[x]|-2}{|[x]|-3}}$ is $(-\infty,a)\cup[b,c)\cup[4,\infty), a<b<c$, then the value of $a+b+c$ is:

[JEE (Main) 2021]

(1) 8 (2)1 (3) -2 (4) -3

MFN051

19. The number of bijective functions $f:\{1,3,5,7,.....99\} \to \{2,4,6,8,.....,100\}$, such that $f(3) \geq f(9) \geq f(15) \geq f(21) \geq ..... \geq f(99)$, is. **[JEE (Main) 2022]**

(1) ${}^{50}P_{17}$ (2) ${}^{50}P_{33}$ (3) $33! \times 17!$ (4) $\frac{50!}{2}$

MFN052

20. Let $f(x)$ be a quadratic polynomial with leading coefficient 1 such that $f(0)=p$, $p \neq 0$ and $f(1)=\frac{1}{3}$. If the equation $f(x)=0$ and $fofofof(x)=0$ have a common real root, then $f(-3)$ is equal to. **[JEE (Main) 2022]**

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MFN053

21. The total number of functions, $f:\{1,2,3,4\} \to \{1,2,3,4,5,6\}$ such that $f(1)+f(2)=f(3)$, is equal to : **[JEE (Main) 2022]**

(1) 60 (2) 90 (3) 108 (4) 126

MFN054

22. Let $f: R \to R$ be a continuous function such that $f(3x)-f(x)=x$. If $f(8)=7$, then $f(14)$ is equal to : **[JEE (Main) 2022]**

(1) 4 (2) 10 (3) 11 (4) 16

MFN055

23. Let $f: \mathbb{R} \to \mathbb{R}$ be defined as $f(x)=x-1$ and $g: \mathbb{R}-\{1,-1\} \to \mathbb{R}$ be defined as $g(x)=\frac{x^2}{x^2-1}$.

Then the function fog is : **[JEE (Main) 2022]**

(1) one-one but not onto function (2) onto but not one-one function

(3) both one-one and onto function (4) neither one-one nor onto function

MFN056

24. The number of one-one function $f:\{a,b,c,d\} \to \{0,1,2,\ldots,10\}$ such that $2f(a)-f(b)+3f(c)+f(d)=0$ is. **[JEE (Main) 2022]**

MFN057

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25. Let $f: N \to R$ be a function such that $f(x+y) = 2f(x)f(y)$ for natural numbers x and y. If $f(1) = 2$, then the value of α for which $\sum_{k=1}^{10} f(\alpha+k) = \frac{512}{3}(2^{20}-1)$ holds, is **[JEE (Main) 2022]**

(1) 2 (2) 3 (3) 4 (4) 6

MFN058

26. Let $f: R \to R$ be a function defined by $f(x) = \left(2\left(1-\frac{x^{25}}{2}\right)\left(2+x^{25}\right)\right)^{\frac{1}{50}}$. If the function $g(x) = f\left(f\left(f(x)\right)\right) + f\left(f(x)\right)$, the greatest integer less than or equal to $g(1)$ is **[JEE (Main) 2022]**

MFN059

27. If domain of the function $\log_e\left(\frac{6x^2+5x+1}{2x-1}\right) + \cos^{-1}\left(\frac{2x^2-3x+4}{3x-5}\right)$ is $(\alpha, \beta) \cup (\gamma, \delta]$, then $18(\alpha^2 + \beta^2 + \gamma^2 + \delta^2)$ is equal to. **[JEE (Main) 2023]**

MFN088

28. Let $R = \{a, b, c, d, e\}$ and $S = \{1, 2, 3, 4\}$. Total number of onto function $f : R \to S$ such that $f(a) \neq 1$, is equal to ____. **[JEE (Main) 2023]**

MFN089

29. Let $f : R - \{0, 1\} \to R$ be a function such that $f(x) + f\left(\frac{1}{1-x}\right) = 1 + x$. Then $f(2)$ is equal to :

[JEE (Main) 2023]

(1) $\frac{9}{2}$ (2) $\frac{9}{4}$ (3) $\frac{7}{4}$ (4) $\frac{7}{3}$

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MFN090

30. Consider a function $f : \mathbb{N} \to \mathbb{R}$, satisfying

$f(1) + 2f(2) + 3f(3) + ... + xf(x) = x(x+1)f(x); x \geq 2$ with $f(1)=1$. Then $\frac{1}{f(2022)} + \frac{1}{f(2028)}$ is equal to **[JEE (Main) 2023]**

(1) 8200 (2) 8000 (3) 8400 (4) 8100

MFN091

31. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \frac{2x}{\sqrt{1+9x^2}}$. If the composition of $f, \underbrace{(f\ o\ f\ o\ f\ o ....o\ f)}_{10 \text{ times}}(x) = \frac{2^{10}x}{\sqrt{1+9\alpha x^2}}$, then the value of $\sqrt{3\alpha+1}$ is equal to .....

[JEE (Main) 2024]

MFN141

32. Let $f, g : R \to R$ be defined as : $f(x) = |x-1|$ and $g(x) = \begin{cases} e^x, & x \geq 0 \\ x+1, & x \leq 0 \end{cases}$. Then the function $f(g(x))$ is **[JEE (Main) 2024]**

(1) neither one-one nor onto.
(2) one-one but not onto.
(3) both one-one and onto.
(4) onto but not one-one.

MFN142

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33. If $S = \{a \in R : |2a - 1| = 3[a] + 2\{a\}\}$, where $[t]$ denotes the greatest integer less than or equal to t and $\{t\}$ represents the fractional part of t, then $72\sum_{a \in S} a$ is equal to _____. **[JEE (Main) 2024]**

MFN143

34. Let $[t]$ be the greatest integer less than or equal to t. Let A be the set of al prime factors of 2310 and $f : A \to \mathbb{Z}$ be the function $f(x) = \left[\log_2\left(x^2 + \left[\frac{x^3}{5}\right]\right)\right]$. The number of one-to-one functions from A to the range of f is : **[JEE (Main) 2024]**

(1) 20 (2) 120 (3) 25 (4) 24

MFN144

35. The function $f : N - \{1\} \to N$; defined by $f(n)$ = the highest prime factor of n, is : **[JEE (Main) 2024]**

(1) both one-one and onto
(2) one-one only
(3) onto only
(4) neither one-one nor onto

MFN145

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EXERCISE - JEE (Advanced) PYQ

1. Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in R$. Then the set of all x satisfying $(fogogof)(x) = (gogof)(x)$, where $(fog)(x) = f(g(x))$, is - **[IIT JEE 2011]**

(A) $\pm\sqrt{n\pi}, n \in \{0,1,2,....\}$

(B) $\pm\sqrt{n\pi}, n \in \{1,2,....\}$

(C) $\frac{\pi}{2} + 2n\pi, n \in \{.....,-2,-1,0,1,2,....\}$

(D) $2n\pi, n \in \{.....,-2,-1,0,1,2,....\}$

MFN080

2. The function $f: [0,3] \to [1,29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is : **[IIT JEE 2012]**

(A) one-one and onto

(B) onto but not one-one

(C) one-one but not onto

(D) neither one-one nor onto

MFN081

3. Let $f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to R$ be given by $f(x) = \left(\log(\sec x + \tan x)\right)^3$. Then:- **[JEE (Advanced) 2014]**

(A) $f(x)$ is an odd function

(B) $f(x)$ is a one-one function

(C) $f(x)$ is an onto function

(D) $f(x)$ is an even function

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MFN083

4. Let X be a set with exactly 5 elements and Y be a set with exactly 7 elements. If α is the number of one one functions from X to Y and β is the number of onto functions from Y to X, then the value of $\frac{1}{5!}(\beta - \alpha)$ is_____ **[JEE (Advanced) 2018]**

MFN084

5. If the function $f: \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = |x|(x - \sin x)$, then which of the following statements is TRUE? **[JEE (Advanced) 2020]**

(A) f is one-one, but NOT onto

(B) f is onto, but NOT one-one

(C) f is BOTH one-one and onto

(D) f is NEITHER one-one NOR onto

MFN085

6. Let the function $f: [0,1] \to \mathbb{R}$ be defined by $f(x) = \frac{4^x}{4^x + 2}$

Then the value of $f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + f\left(\frac{3}{40}\right) + ... + f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right)$ is_____

[JEE (Advanced) 2020]

MFN086

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7. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by

$$f(x)=\begin{cases} x^2 \sin\left(\frac{\pi}{x^2}\right), & \text{if } x \neq 0, \\ 0, & \text{if } x = 0. \end{cases}$$

Then which of the following statements is TRUE ? **[JEE (Advanced) 2024]**

(A) $f(x) = 0$ has infinitely many solutions in the interval $\left[\frac{1}{10^{10}}, \infty\right)$.

(B) $f(x) = 0$ has no solutions in the interval $\left[\frac{1}{\pi}, \infty\right)$.

(C) The set of solutions of $f(x) = 0$ in the interval $\left(0, \frac{1}{10^{10}}\right)$ is finite.

(D) $f(x) = 0$ has more than 25 solutions in the interval $\left(\frac{1}{\pi^2}, \frac{1}{\pi}\right)$.

MFN146

8. Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$, and $g : \mathbb{R} \to (0, \infty)$ be a function such that $g(x + y) = g(x)\, g(y)$ for all $x, y \in \mathbb{R}$. If $f\left(\frac{-3}{5}\right) = 12$ and $g\left(\frac{-1}{3}\right) = 2$, then the value of $\left(f\left(\frac{1}{4}\right) + g(-2) - 8\right) g(0)$ is _____. **[JEE (Advanced) 2024]**

MFN147

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	D	B	B	D	A	C	D	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	B	A	D	C	C	A	D	C	B

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C	A,C	A,B,C	B,C,D	C,D	A,B,C,D	A,B,C	A,D	B,C	A,D
Que.	11	12	13	14	15	16	17	18		
Ans.	A,B,C	A	C	A	D	D	C	A		
Que.	19									
Ans.	A → P,S ; B → R ; C → Q,S ; D → T									

EXERCISE (S)

1. 6 **2.** 1002.50 **3.** 2 **4.** 11 **5.** 6
6. 5 **7.** 6 **8.** 401 **9.** 11 **10.** 15

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	2	1	3	2	4	2	3	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	5.00	4	26	3	3	3	2	25
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	2	4	31	3	2	20	180	2	4
Que.	31	32	33	34	35					
Ans.	1024	1	18	2	4					

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8
Ans.	A	B	A,B,C	119	C	19.00	D	51

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